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CONTROL DEVICE, MACHINE TOOL, AND CONTROL METHOD

Abstract

A control device for a machine tool comprises: an acquisition unit that acquires the type of a tool used for machining a machining object; a determination unit that determines, on the basis of the type of the tool, whether the operation of a mist collector is permitted or prohibited; and a collector control unit that turns ON the mist collector if it is determined that the operation of the mist collector is permitted, and turns OFF the mist collector if it is determined that the operation of the mist collector is prohibited.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a control device for controlling a machine tool including a mist collector, a machine tool related to the control device, and a control method.

BACKGROUND ART

[0002] A mist collector collects mist generated in a machining area of a machine tool (see also JP 2012-076006 A). The mist collector collects the mist within the machining area, and prevents the mist from leaking out to the exterior of the machining area. The mist includes vaporized coolant.

SUMMARY OF THE INVENTION

[0003] In the mist collector, vibration may occur during operation of the mist collector. In this case, there is a concern that the vibration generated by the mist collector affects the machining on a workpiece and the machine accuracy is reduced.

[0004] An object of the present invention is to solve the aforementioned problem.

[0005] A first aspect of the present invention is a control device for a machine tool including a mist collector configured to collect mist in a machining area, the machine tool being configured to machine a workpiece within the machining area, the control device including an acquisition unit configured to acquire a type of a tool used for performing machining on the workpiece, a determination unit configured to determine whether to permit or prohibit an operation of the mist collector based on the type of the tool, and a collector control unit configured to turn ON the operation of the mist collector in a case that a determination has been made to permit the operation of the mist collector, and turn OFF the operation of the mist collector in a case that a determination has been made to prohibit the operation of the mist collector.

[0006] A second aspect of the present invention is a machine tool including the control device according to the first aspect.

[0007] A third aspect of the present invention is a control method for controlling by a computer a machine tool including a mist collector configured to collect mist in a machining area, the machine tool being configured to machine a workpiece within the machining area, the control method including an acquisition step of acquiring a type of a tool used for performing machining on the workpiece, a determination step of determining whether to permit or prohibit an operation of the mist collector based on the type of the tool, and a collector control step of turning ON the operation of the mist collector in a case that a determination has been made to permit the operation of the mist collector, and turning OFF the operation of the mist collector in a case that a determination has been made to prohibit the operation of the mist collector.

[0008] According to the present invention, for example, in the case that a workpiece is machined by a tool used for precision machining, the collection of mist by the mist collector can be stopped. Therefore, it is possible to prevent the vibration generated by the mist collector from affecting the machining on the workpiece. As a result, it is possible to suppress degradation of the machine accuracy.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. **1** is a schematic diagram of a machine tool according to an embodiment;

[0010] FIG. **2** is a block diagram of a control device;

[0011] FIG. **3** is a flowchart illustrating a control method according to the embodiment;

[0012] FIG. **4** is a schematic diagram of a machine tool according to an exemplary modification **1**; and

[0013] FIG. **5** is a block diagram of a control device according to an exemplary modification **2**.

DETAILED DESCRIPTION OF THE INVENTION

Embodiment

[0014] FIG. **1** is a schematic diagram of a machine tool **10** according to an embodiment.
[0015] The X direction and the Y direction shown in FIG. **1** are directions that are parallel to a horizontal plane. The X direction and the Y direction are mutually orthogonal with each other. The Z direction shown in FIG. **1** is a direction that is parallel to the direction of gravity. Accordingly, the Z direction is perpendicular to the X direction and the Y direction. However, the Z direction shown in FIG. **1** indicates a direction that is opposite to the direction of gravity.
[0016] The machine tool **10** includes a machining device **12** and a control device **14**.
[0017] The machining device **12** is a machine device that carries out machining on a workpiece using a tool **16**. The machining device **12** is equipped with a spindle **18**, a spindle head **20**, a column **22**, a pedestal **24**, a table **26**, a table drive unit **28**, a cover **30**, a coolant supply device **32**, and a mist collector **34**.
[0018] A tool holder **36** is attached to the spindle **18**. The tool holder **36** is capable of being attached and detached to and from the spindle **18**. The tool holder **36** retains the tool **16**. The tool **16**, for example, is a spring-necked turning tool, a drill, an end mill, a milling cutter, or the like.
[0019] The machining device **12** is further equipped with a tool magazine **38**. The tool magazine **38** detachably retains a plurality of the tools **16**. One of the plurality of the tools **16** that are retained in the tool magazine **38** is changeably attached to the tool holder **36**.
[0020] The spindle head **20** supports the spindle **18**. Further, the spindle head **20** includes a motor that causes the spindle **18** to be rotated. The tool **16**, which is attached to the spindle **18** via the tool holder **36**, rotates together with the spindle **18**.
[0021] The column **22** supports the spindle head **20**. Further, the column **22** includes a motor that causes the spindle head **20** to be moved in the Z direction. The column **22** is supported on the pedestal **24**.
[0022] The pedestal **24** is installed on an installation surface. The installation surface, for example, is a floor of a factory. The installation surface may be a support surface of a platform that is provided on the floor. The installation surface extends, for example, parallel to the horizontal plane. The pedestal **24** may be equipped with a plurality of leg members **24***a*. Each of the leg members **24***a*, for example, may be a caster, a jack, or the like.
[0023] The table drive unit **28** is supported on the pedestal **24**. The table drive unit **28** is equipped with a first slide unit **42**, a saddle **44**, and a second slide unit **46**.
[0024] The first slide unit **42** is installed on the pedestal **24**. The first slide unit **42** includes, for example, guide rails that extend in the Y direction. The first slide unit **42** supports the saddle **44**.
[0025] The saddle **44** moves in the Y direction in accordance with the driving of a non-illustrated motor. The motor is controlled by the control device **14**. The saddle **44** moves while being guided by the first slide unit **42**.
[0026] The second slide unit **46** is mounted on the saddle **44**. The second slide unit **46** includes, for example, guide rails that extend in the X direction.
[0027] The table **26** supports a non-illustrated workpiece downwardly of the spindle **18**. The table **26** is supported on the second slide unit **46**. The table **26** moves in the X direction in accordance with the driving of a non-illustrated motor. The motor is controlled by the control device **14**. The table **26** moves while being guided by the second slide unit **46**.
[0028] The cover **30** covers the spindle **18**, the spindle head **20**, the column **22**, the pedestal **24**, the table **26**, and the table drive unit **28**. Consequently, the cover **30** forms a machining area **48**. A workpiece is machined within the machining area **48**.
[0029] The cover **30** is further equipped with a non-illustrated door and a non-illustrated window. The operator can carry out an introduction operation of the workpiece into the machining area **48** through the door that is in an opened state. Further, the operator can easily confirm the condition within the machining area **48** via the window.
[0030] The coolant supply device **32** is a device that supplies the coolant to the machining area **48**. The coolant supply device **32** is equipped with a coolant tank **50**, a nozzle **52**, a supply pipe **54**, and

a pump **56**.

[0031] The coolant tank **50** is a tank for storing coolant. The coolant tank **50** is installed externally of the machining area **48**.

[0032] The nozzle **52** is a discharge unit that discharges the coolant. The nozzle **52** is disposed within the machining area **48**. Moreover, the coolant supply device **32** may be equipped with a plurality of the nozzles **52**.

[0033] The supply pipe **54** is a pipe that connects the coolant tank **50** and the nozzle **52**. Moreover, the coolant supply device **32** may be equipped with a plurality of the supply pipes **54**. The number of the supply pipes **54** is determined, for example, in accordance with the number of the nozzles **52**. The supply pipe **54** passes through the cover **30**, and connects the coolant tank **50** and the nozzle **52**.

[0034] The pump **56** is connected to the supply pipe **54**. The pump **56** draws in the coolant within the coolant tank **50**, and delivers the coolant to the nozzle **52**. Consequently, the coolant is discharged from the nozzle **52** into the machining area **48**. The pump **56** is controlled by the control device **14**.

[0035] The coolant that is discharged into the machining area **48** cools the tool **16** and the workpiece. The coolant discharged to the machining area **48** is vaporized, and the vaporized mist is generated in the machining area **48**.

[0036] The mist collector **34** is a device that serves to collect the mist within the machining area **48**. The mist collector **34** is installed externally of the machining area **48**. Further, the mist collector **34** is connected to the cover **30** via a duct **58**. The mist collector **34** collects the mist by drawing in air within the machining area **48**. In accordance with this feature, the mist is prevented from leaking out to the exterior of the machining area **48** via small gaps that occur in the machining device **12**.

[0037] In the case that a workpiece is cut by the tool **16**, fine or minute cutting chips are generated in the form of powdery dust within the machining area **48**. In the case that the mist collector **34** draws in air within the machining area **48**, not only the mist but also the powdery dust is collected by the mist collector **34**. In accordance with this feature, the powdery dust is also prevented from leaking out to the exterior of the machining area **48** via small gaps that occur in the machining device **12**.

[0038] The mist collector **34** may be connected to the coolant tank **50**. Consequently, the mist that is collected by the mist collector **34** can be returned, as the coolant, to the coolant tank **50**.

[0039] In the case that the mist collector **34** and the coolant tank **50** are connected to each other, it is preferable for the mist collector **34** and the coolant tank **50** to be connected to each other via a non-illustrated filtering device (filter). The filtering device removes impurities from the coolant that is delivered from the mist collector **34** to the coolant tank **50**. By connecting the mist collector **34** and the coolant tank **50** via the filtering device, a clean coolant can be returned from the mist collector **34** to the coolant tank **50**. Impurities in the coolant, for example, are the cutting chips that have been collected together with the mist.

[0040] FIG. **2** is a block diagram of the control device **14**.

[0041] The control device **14** is a computer that controls the machining device **12**. The control device **14**, for example, is a numerical control device. The control device **14** is equipped with a display unit **60**, an operation unit **62**, a storage unit **64**, a computation unit **66**, and a standby electrical power supply unit **68**.

[0042] The display unit **60** is a display device equipped with a display screen **60***d*. The display unit **60**, for example, is a liquid crystal display device or an OEL (Organic Electro-Luminescence) display device.

[0043] The operation unit **62** is an input device that receives instructions from the operator to the control device **14**. The operation unit **62** includes, for example, an operation panel **62***a*, and a touch panel **62***b* or the like. The touch panel **62***b* is provided on the display screen **60***d*. The operation unit

62 (the operation panel **62***a*) may be equipped with a keyboard, a mouse, or the like.
[0044] The storage unit **64** is constituted by a non-illustrated volatile memory, and a non-illustrated nonvolatile memory. As an example of the volatile memory, there may be cited a random access memory (RAM) or the like. As an example of the nonvolatile memory, there may be cited a ROM (Read Only Memory) and a flash memory or the like. Data and the like may be stored, for example, in the volatile memory. Programs, tables, maps, etc. may be stored, for example, in the nonvolatile memory. At least a portion of the storage unit **64** may be included in a processor, an integrated circuit, or the like. In the present embodiment, the storage unit **64** stores a control program **70** and a machining program **72**.
[0045] The control program **70** is a program for the purpose of causing the control device **14** to execute the control method according to the present embodiment. The control method will be described later.
[0046] The machining program **72** is a program including control commands for the machining device **12**. The machining program **72** includes, for example, a plurality of control commands for the purpose of controlling each of the aforementioned motors. Further, the machining program **72** also includes, for example, a plurality of control commands for the purpose of controlling the coolant supply device **32**. The machining program **72** is created or edited in advance by the operator.
[0047] The computation unit **66** is constituted by a processor including, for example, a CPU (Central Processing Unit), and a GPU (Graphics Processing Unit) or the like. The computation unit **66** includes a processing circuitry.
[0048] The computation unit **66** includes a machining control unit **74**, an acquisition unit **76**, a determination unit **78**, and a collector control unit **80**. The machining control unit **74**, the acquisition unit **76**, the determination unit **78**, and the collector control unit **80** are realized by the processor of the computation unit **66** executing the control program **70**. At least a portion of the machining control unit **74**, the acquisition unit **76**, the determination unit **78**, and the collector control unit **80** may be realized by an integrated circuit such as an ASIC (Application Specific Integrated Circuit) or an FPGA (Field-Programmable Gate Array). At least a portion of the machining control unit **74**, the acquisition unit **76**, the determination unit **78**, and the collector control unit **80** may be configured by an electronic circuit including a discrete device.
[0049] The machining control unit **74**, by controlling the machining device **12** based on the machining program **72**, carries out machining on the workpiece. The machining control unit **74** controls the mounting of the tool **16**, the rotation of the spindle **18**, the movement of the spindle head **20**, the movement of the table **26**, and the like. However, the mist collector **34** of the machining device **12** is controlled by the collector control unit **80**.
[0050] The acquisition unit **76** acquires a type of the tool **16** used for machining a workpiece. The acquisition unit **76** starts to acquire the type of the tool **16** in the case that the mounting of the tool **16** is detected. The mounting of the tool **16** is performed before the machining of the workpiece is started. The mounting of the tool **16** includes a case that the tool **16** is changed with a new tool **16**. The detection of the mounting of the tool **16** is performed by a switch, a sensor, or the like provided in the spindle head **20**. In this case, the acquisition unit **76** recognizes the case that the mounting of the tool **16** is detected based on the detection signal output from the switch, the sensor, or the like, and starts to acquire the type of the tool **16**.
[0051] The acquisition unit **76** may acquire the type of the tool **16** based on the tool identification information. In this case, a tool table stored in the storage unit **64** is used. In the tool table, the tool identification information and the type information indicating the type of the tool **16** are associated with each other. The tool identification information is information for identifying the tool **16** and includes at least one of a tool number, a tool weight, a tool diameter, a tool length, and a tool run-out degree. The tool run-out degree is a run-out degree (run-out width from the rotary axis) of the tool **16** rotating at a predetermined rotation speed.

[0052] For example, the acquisition unit **76** can acquire the type of the tool **16** corresponding to the tool number included in the machining program **72** by using the tool table. For example, the acquisition unit **76** can acquire the type of the tool **16** corresponding to at least one of the tool weight, the tool diameter, the tool length, and the tool run-out degree detected by a detection mechanism, using the tool table. The detection mechanism includes a camera, a measurement instrument, a sensor, and the like attached to the machining device **12**. Examples of the sensor include a weight sensor and a vibration sensor.
[0053] The acquisition unit **76** may acquire the type of the tool **16** input from the operation unit **62** in accordance with an operation by an operator. In this case, the acquisition unit **76** may control the display unit **60** or the like to notify the operator that the type of the tool **16** should be input.
[0054] The determination unit **78** determines whether to permit or prohibit the operation of the mist collector **34** based on the type of the tool **16** acquired by the acquisition unit **76**. In this case, a determination table TB stored in the storage unit **64** is used. In the determination table TB, type information indicating the type of the tool **16** and state information indicating whether the operation of the mist collector **34** is permitted or prohibited are associated with each other. For example, when the type information indicates a milling cutter or a drill, the determination unit **78** determines that the operation of the mist collector **34** is permitted. When the type information indicates a spring-necked turning tool or an end mill, the determination unit **78** determines that the operation of the mist collector **34** is prohibited.
[0055] When the type information indicates the type of the tool **16** used for precision machining, the determination unit **78** may determine to prohibit the operation of the mist collector **34**. When the type information indicates the type of the tool **16** having a natural frequency that matches or approximates the vibration of the mist collector **34**, the determination unit **78** may determine to prohibit the operation of the mist collector **34**.
[0056] The collector control unit **80** controls the mist collector **34** based on the determination result of the determination unit **78**. In the case that the determination unit **78** has determined to permit the operation of the mist collector **34**, the collector control unit **80** turns ON the operation of the mist collector **34**. In this case, the mist collector **34** is automatically started from the timing when the operation of the mist collector **34** is turned ON, and starts the collection of the mist.
[0057] On the other hand, in the case that the determination unit **78** determines that the operation of the mist collector **34** is prohibited, the collector control unit **80** turns OFF the operation of the mist collector **34**. In this case, the mist collector **34** is not started.
[0058] The standby electrical power supply unit **68** is an electrical power supply that is separate from the main power supply of the control device **14**. The standby electrical power supply unit **68** includes, for example, a battery. The standby electrical power supply unit **68** is integrated in the control device **14**. However, the standby electrical power supply unit **68** may be provided in the machine tool **10** as an external power supply for the control device **14**. Moreover, it should be noted that the main power supply for the control device **14** is not shown in the drawings.
[0059] In the case that the main power supply of the control device **14** is turned OFF while the mist collector **34** is in operation, the standby electrical power supply unit **68** supplies electrical power to each of the components of the control device **14**. In accordance with this feature, even after the main power supply has been turned OFF, the collector control unit **80** can continue to control the mist collector **34**.
[0060] If electrical power is supplied to the mist collector **34** from a power supply other than the main power supply of the control device **14**, the mist collector **34** can continue to operate even when the main power supply of the control device **14** is turned OFF. In such a case, the main power supply of the control device **14** may be turned OFF before the collector control unit **80** stops the mist collector **34**. In such a case, by supplying electrical power from the standby electrical power supply unit **68**, the collector control unit **80** can cause the mist collector **34** to be automatically stopped even after the main power supply of the control device **14** is turned OFF. In accordance

with this feature, the wasteful power consumption of the mist collector **34** is suppressed.

[0061] FIG. **3** is a flowchart illustrating the control method according to the embodiment.

[0062] The control device **14** can execute a control method, for example, as illustrated in FIG. **3**. The control method of FIG. **3** includes an acquisition step S**1**, a determination step S**2**, a collector control step S**3**, and a machining step S**4**. The collector control step S**3** includes an operation ON step S**31** and an operation OFF step S**32**. The acquisition step S**1**, the determination step S**2**, and the collector control step S**3** are executed before the machining on the workpiece is started.

[0063] The acquisition step S**1** is a step of acquiring the type of the tool **16**. In the acquisition step S**1** according to the present embodiment, the type of the tool **16** is acquired by the acquisition unit **76**. When the mounting of the tool **16** to the spindle **18** is detected, the acquisition unit **76** analyzes the machining program **72** by which the mounting of the tool **16** has been executed and acquires the type of the tool **16**.

[0064] The determination step S**2** is a step of determining whether to permit or prohibit the operation of the mist collector **34** based on the type of the tool **16** acquired in the acquisition step S**1**. In the determination step S**2** according to the present embodiment, whether the operation of the mist collector **34** is permitted or prohibited by the determination unit **78** is determined by the determination unit **78** based on the determination table TB.

[0065] If the state information of the determination table TB corresponding to the type of the tool **16** acquired in the acquisition step S**1** indicates permission of the operation of the mist collector **34** (S**2**: NO), the determination unit **78** determines that the operation of the mist collector **34** is permitted. Conversely, if the state information of the determination table TB corresponding to the type of the tool **16** acquired in the acquisition step S**1** indicates prohibition of the operation of the mist collector **34** (S**2**: YES), the determination unit **78** determines that the operation of the mist collector **34** is prohibited. Depending on the determination content in the determination step S**2**, the operation ON step S**31** or the operation OFF step S**32** is started.

[0066] The operation ON step S**31** is a step in which the mist collector **34** is turned ON to operate. In the operation ON step S**31** according to the present embodiment, the collector control unit **80** turns ON the operation of the mist collector **34**. In this case, the machining of the workpiece is started after the collection of the mist by the mist collector **34** is started.

[0067] The operation OFF step S**32** is a step in which the mist collector **34** is turned OFF so as not to operate. In the operation OFF step S**32** according to the present embodiment, the collector control unit **80** turns OFF the operation of the mist collector **34**. In this case, the machining on the workpiece is started without collecting the mist by the mist collector **34**.

[0068] The machining step S**4** is a step of machining a workpiece. In the machining step S**4** according to the present embodiment, the workpiece is machined under the control of the machining control unit **74** to the machining device **12**. In this case, the machining control unit **74** machines the workpiece based on the machining program **72** by which the mounting of the tool **16** to the spindle **18** is executed.

[0069] If the operation of the mist collector **34** is turned OFF in the collector control step S**3**, the control method of FIG. **3** is brought to an end when the machining on the workpiece is completed. If the operation of the mist collector **34** is turned ON in the collector control step S**3**, the collector control unit **80** stops the mist collector **34** at the timing when the machining on the workpiece is completed. In this case, when the stopping of the mist collector **34** is completed, the control method of FIG. **3** is completed.

[0070] As described above, in the present embodiment, the type of the tool **16** used for machining the workpiece is acquired, and it is determined whether the operation of the mist collector **34** is permitted or prohibited based on the type of the tool **16**. If the determination is made to permit the operation of the mist collector **34**, the operation by the mist collector **34** is turned ON, and if the determination is made to prohibit the operation of the mist collector **34** is made, the operation by the mist collector **34** is turned OFF.

[0071] In accordance with such features, for example, in the case that a workpiece is machined by a tool **16** used for precision machining, the collection of mist by the mist collector **34** can be stopped. Therefore, it is possible to prevent the vibration generated by the mist collector **34** from affecting the machining on the workpiece. As a result, it is possible to suppress a reduction in the machine accuracy. In addition, since the mist collector **34** is stopped, the electric power consumption of the mist collector **34** is suppressed. Further, the operation of the mist collector **34** can be turned ON and OFF regardless of the coolant discharge state.

EXEMPLARY MODIFICATIONS

[0072] Hereinafter, a description will be given concerning exemplary modifications of the above-described embodiment. However, any descriptions that are duplicative or overlap with those of the above-described embodiment will be appropriately omitted in the following description. Unless otherwise specified, the same reference numerals as in the above-described embodiment are used in referring to the elements that have already been described in the embodiment.

Exemplary Modification 1

[0073] FIG. **4** is a schematic diagram of a machine tool **101** (**10**) according to an exemplary modification **1**. The machine tool **101** is further equipped with a sub-control device **82**. In the machine tool **101**, the standby electrical power supply unit **68** of the control device **14** may be omitted.

[0074] The sub-control device **82** is a computer that is separate from the control device **14**. The sub-control device **82**, for example, is equipped with a processor and a memory. The sub-control device **82** may also be equipped with an integrated circuit, a discrete device, or the like.

[0075] In the case that the control device **14** is stopped, the sub-control device **82** controls the mist collector **34** instead of the collector control unit **80**. Therefore, even if the control device **14** is stopped, the mist collector **34** is controlled by the sub-control device **82** in the same manner as in the embodiment.

[0076] For example, in the case that the machining is completed, the operator issues an instruction to the control device **14** to immediately stop the machining. Consequently, the control device **14** immediately stops the machining after the completion of the machining. However, as noted previously, it is preferable for the mist collector **34** to collect the mist until the predetermined time period has elapsed from the completion of the machining. In such a case, the sub-control device **82** is capable of controlling the mist collector **34** instead of the control device **14**.

[0077] For example, if the main power supply of the control device **14** is turned OFF before the collector control unit **80** causes the mist collector **34** to be stopped, the sub-control device **82** can control the mist collector **34**, instead of the control device **14**.

[0078] Moreover, it is preferable for the sub-control device **82** and the control device **14** to communicate with each other as appropriate, and to share the data necessary for controlling the mist collector **34**. For example, the sub-control device **82** and the control device **14** share the type of the tool **16** acquired by the acquisition unit **76**, the determination content by the determination unit **78**, or the progress of the machining. In accordance with these features, the sub-control device **82** is capable of smoothly taking over the control that was being performed by the collector control unit **80**. According to the present exemplary modification, even after the control device **14** is stopped, the control of the mist collector **34** can be continued by the sub-control device **82**.

Exemplary Modification 2

[0079] FIG. **5** is a block diagram of a control device **142** (**14**) according to an exemplary modification 2. The control device **142** is further equipped with an alarm output unit **84**.

[0080] The alarm output unit **84** outputs an alarm in the case that an abnormality has occurred in the machine tool **10**. For example, the machine tool **10** is appropriately provided with non-illustrated sensors for the purpose of detecting malfunctions or troubles in each of the respective components such as the spindle **18**, the spindle head **20**, the table drive unit **28**, etc. The alarm output unit **84** determines whether or not a malfunction has occurred in the machine tool **10** based on the signals output by the sensors. In the case that a malfunction is detected in any of the respective components of the machine tool **10**, the alarm output unit **84** issues a notification to the operator, for example via the display unit **60**, to the effect that a malfunction has occurred.

[0081] In the case that the alarm output unit **84** has output an alarm prior to the start of the machining, the machining control unit **74** does not start the machining until the cause of the alarm is eliminated. Further, in the case that the alarm output unit **84** has output an alarm after the start of the machining, the machining control unit **74** suspends the machining based on the machining program **72** until the cause of the alarm is eliminated.

[0082] On the other hand, in the case that an alarm is output and the operation of the mist collector **34** is ON, the collector control unit **80** stops the operation of the mist collector **34** until the cause of the alarm is eliminated. In accordance with such features, it is possible to prevent the mist collector **34** from continuing to operate in case of emergency of the machine tool **10**. However, in the case that an operator instructs to start operation at the time of abnormal stop of the machining device **12**, the collector control unit **80** may restart the operation of the mist collector **34**.

Combination of Plural Exemplary Modifications

[0083] The above-described plurality of exemplary modifications may be combined as appropriate within a range in which there are no contradictions therebetween.

Modified Embodiments

[0084] It should be noted that the present invention is not limited to the embodiment described above, and various alternative or additional configurations may be adopted therein without departing from the essence and gist of the present invention.

[0085] For example, according to the above embodiment, the mist collector **34** stops when the machining control unit **74** completes the machining. However, the collector control unit **80** may control the mist collector **34** to cause the mist collector **34** to collect the mist in the machining area **48** until a predetermined time period has elapsed from the completion of the machining. By intentionally operating the mist collector **34** after the completion of the machining, it is possible to prevent a loss in the collection of the mist. The predetermined time period is specified in advance to the collector control unit **80** by, for example, an operator through the operation unit **62**. However, the predetermined time period may be specified by the manufacturer of the machine tool **10**.

[0086] In addition, for example, according to the above-described embodiment, the collector control unit **80** causes the mist collector **34** to start to collect the mist before the machining on the workpiece is started. However, the collector control unit **80** may cause the mist collector **34** to start to collect the mist after a predetermined stand-by time has elapsed from the start of the machining of the workpiece. By intentionally operating the mist collector **34** after the start of the machining, it is possible to suppress the electric power consumption of the mist collector **34**. The stand-by time is specified in advance to the collector control unit **80** by, for example, an operator through the operation unit **62**. However, the stand-by time may be specified by the manufacturer of the machine tool **10**.

[0087] For example, the collector control unit **80** may receive instructions from the operator to permit the operation of the mist collector **34**, during a period from the time when the determination is made to prohibit the operation of the mist collector **34** to the time when the machining on the workpiece is completed. When the instructions to permit the operation of the mist collector **34** are issued, the collector control unit **80** turns ON the operation of the mist collector **34** even if the determination to prohibit the operation has been made. Similarly, the collector control unit **80** may receive instructions from the operator to prohibit the operation of the mist collector **34**, during a period from the time when the determination is made to permit the operation of the mist collector **34** to the time when the machining on the workpiece is completed. When the instructions to prohibit the operation of the mist collector **34** are issued, the collector control unit **80** turns OFF the operation of the mist collector **34** even if the determination to permit the operation has been made.

In this way, when the collector control unit **80** receives the instructions from the operator to permit or prohibit the operation of the mist collector **34**, the collector control unit **80** gives priority to the instructions and turns ON or OFF the operation of the mist collector **34**, regardless of the determination by the determination unit **78**. In accordance with this feature, it is possible to turn ON or OFF the operation of the mist collector **34** with priority given to the operator's intention.

[0088] For example, the determination unit **78** may determine whether to permit or prohibit the operation of the mist collector **34** based on the type of the tool **16** and the material of the workpiece. The material of the workpiece may be added to the determination table TB or may be input by an operator. By adding the material of the workpiece to the determination conditions of the determination unit **78**, it is possible to define in detail the case in which the operation of the mist collector **34** is permitted and the case in which the operation of the mist collector **34** is prohibited. The vibration generated while the workpiece is being machined (machining vibration) may resonate with the vibration generated by the mist collector **34** (collector vibration). The machining vibration depends not only on the tool **16** but also on the material of the workpiece. Therefore, by adding the material of the workpiece to the determination conditions of the determination unit **78**, it is possible to appropriately determine to prohibit the operation of the mist collector **34** in the case where there is a possibility that the collector vibration and the machining vibration resonate.

[0089] For example, the machining control unit **74** may control the pump **56** of the coolant in accordance with the operating state of the mist collector **34**. That is, if the determination unit **78** determines to permit the operation of the mist collector **34**, the machining control unit **74** drives the pump **56** of the coolant. On the other hand, if the determination unit **78** determines to prohibit the operation of the mist collector **34**, the machining control unit **74** does not drive the pump **56** of the coolant. In this way, it is possible to prevent the vibration generated by the coolant supply device **32** from affecting the machining on the workpiece.

[0090] Also, for example, the state information may be optionally defined in the machining program **72**. In this case, the determination unit **78** can determine whether to permit or prohibit the operation of the mist collector **34** based on the machining program **72**. Therefore, the determination table TB need not be stored in the storage unit **64**. Therefore, it is possible to reduce the storage capacity of the storage unit **64**.

[0091] Further, for example, the coolant discharge method is not limited to being that of the embodiment. For example, the coolant may be discharged using a center-through method. In that case, the coolant supply device **32** supplies the coolant to the spindle **18**. Further, the coolant may flow along the inner wall of the cover **30** (the machining area **48**).

[0092] For example, the machining device **12** may further be equipped with a non-illustrated recovery member in order to recover the coolant that falls downwardly of the table **26**. Such a recovery member, for example, is an oil pan provided on the pedestal **24**. A portion of the coolant supplied to the machining area **48** falls downwardly of the table **26** without turning into a mist. According to the present exemplary modification, it is possible to collect the coolant that has fallen downwardly of the table **26**. The recovered coolant may be returned to the coolant tank **50**. In accordance with this feature, the coolant supply device **32** is capable of reusing the coolant that has been collected. In this instance, it is preferable for a filtering device (a filter) to be disposed between the recovery member and the coolant tank **50**. In accordance with this feature, a clean coolant can be returned to the coolant tank **50**.

Invention Obtained from the Embodiment

[0093] Invention that can be grasped from the above-described embodiment and the exemplary modifications thereof will be described below.

[0094] (1) The first aspect of invention is the control device (**14**) for the machine tool (**10**) including the mist collector (**34**) configured to collect mist in the machining area (**48**), the machine tool being configured to machine the workpiece within the machining area, the control device including the acquisition unit (**76**) configured to acquire the type of the tool (**16**) used for

performing machining on the workpiece, the determination unit (**78**) configured to determine whether to permit or prohibit the operation of the mist collector based on the type of the tool, and the collector control unit (**80**) configured to turn ON the operation of the mist collector in the case that the determination has been made to permit the operation of the mist collector, and turn OFF the operation of the mist collector in the case that the determination has been made to prohibit the operation of the mist collector.
[0095] In accordance with such features, for example, in the case that the workpiece is machined by the tool used for precision machining, the collection of mist by the mist collector can be stopped. Therefore, it is possible to prevent the vibration generated by the mist collector from affecting the machining on the workpiece. As a result, it is possible to suppress degradation of the machine accuracy.
[0096] (2) The first aspect of the present invention is the control device, wherein the collector control unit may use the determination table (TB) in which type information indicating the type of the tool and state information indicating whether to permit or prohibit the operation of the mist collector are associated with each other. In accordance with this feature, it is possible to manage the tool for turning ON or OFF the operation of the mist collector.
[0097] (3) The first aspect of the present invention is the control apparatus, wherein the collector control unit may turn OFF the operation of the mist collector in the case that the type of the tool used for precision machining has been acquired. In accordance with this feature, it is possible to suppress degradation of the machine accuracy in the precision machining.
[0098] (4) The first aspect of the present invention is the control device, wherein the acquisition unit may start to acquire the type of the tool that is mounted in the case that mounting of the tool is detected. In accordance with this feature, even if the tool is changed, it is possible to turn ON or OFF the operation of the mist collector based on the type of tool after being changed.
[0099] (5) The first aspect of the present invention is the control device, wherein the collector control unit may determine whether to permit or prohibit the operation of the mist collector before the machining of the workpiece is started. In accordance with this feature, in the case that the machining is to be performed using the type of the tool with which the operation of the mist collector is prohibited, it is possible to prevent beforehand the operation of the mist collector from being turned ON before the machining is performed.
[0100] (6) The first aspect of the present invention is the control device, wherein in the case that the determination has been made to permit the operation of the mist collector, the collector control unit may stop driving the mist collector after the predetermined time period has elapsed from completion of the machining on the workpiece. In accordance with this feature, the amount of the mist that remains in the machining area after the completion of the machining can be reduced, while the electrical power consumption of the mist collector can be prevented from becoming excessively large.
[0101] (7) The first aspect of the present invention is the control device, which may further include the alarm output unit (**84**) configured to output the alarm in the case that the abnormality has occurred in the machine tool, wherein, in the case that the alarm is output while the operation of the mist collector is ON, the collector control unit may stop the operation of the mist collector. In accordance with such features, it is possible to prevent the mist collector from continuing to operate in case of emergency of the machine tool.
[0102] (8) A second invention is a machine tool (**10**) including the control device according to the first aspect of invention. Since the control device according to the first aspect of the present invention is included, it is possible to suppress degradation of the machine accuracy.
[0103] (**9**) The second aspect of the present invention is the machine tool, which may further include the sub-control device (**82**) configured to control the mist collector instead of the collector control unit in the case that the control device is stopped. In accordance with this feature, even if the control device is stopped, the mist collector can be controlled.

[0104] (**10**) A third aspect of invention is the control method for controlling by a computer (**14**) the machine tool including the mist collector configured to collect mist in the machining area, the machine tool being configured to machine the workpiece within the machining area, the control method including the acquisition step (S**1**) of acquiring the type of the tool used for performing machining on the workpiece, the determination step (S**2**) of determining whether to permit or prohibit the operation of the mist collector based on the type of the tool, and the collector control step (S**3**) of turning ON the operation of the mist collector in the case that the determination has been made to permit the operation of the mist collector, and turning OFF the operation of the mist collector in the case that the determination has been made to prohibit the operation of the mist collector.
[0105] In accordance with such features, for example, in the case that the workpiece is machined by the tool used for precision machining, the collection of mist by the mist collector can be stopped. Therefore, it is possible to prevent the vibration generated by the mist collector from affecting the machining on the workpiece. As a result, it is possible to suppress degradation of the machine accuracy.
REFERENCE SIGNS LIST
[0106] **10**, **101**: machine tool [0107] **12**: machining device [0108] **14**, **142**: control device [0109] **32**: coolant supply device [0110] **34**: mist collector [0111] **48**: machining area [0112] **72**: machining program [0113] **74**: machining control unit [0114] **76**: acquisition unit [0115] **78**: determination unit [0116] **80**: collector control unit [0117] **82**: sub-control device [0118] **84**: alarm output unit [0119] TB: determination table

Claims

1. A control device for a machine tool comprising a mist collector configured to collect mist in a machining area, the machine tool being configured to machine a workpiece within the machining area, the control device comprising: an acquisition unit configured to acquire a type of a tool used for performing machining on the workpiece; a determination unit configured to determine whether to permit or prohibit an operation of the mist collector based on the type of the tool; and a collector control unit configured to turn ON the operation of the mist collector in a case that a determination has been made to permit the operation of the mist collector, and turn OFF the operation of the mist collector in a case that a determination has been made to prohibit the operation of the mist collector.
2. The control device according to claim 1, wherein the collector control unit uses a determination table in which type information indicating the type of the tool and state information indicating whether to permit or prohibit the operation of the mist collector are associated with each other.
3. The control device according to claim 1, wherein the collector control unit turns OFF the operation of the mist collector in a case that the type of the tool used for precision machining has been acquired.
4. The control device according to claim 1, wherein the acquisition unit starts to acquire the type of the tool that is mounted in a case that mounting of the tool is detected.
5. The control device according to claim 1, wherein the collector control unit determines whether to permit or prohibit the operation of the mist collector before the machining of the workpiece is started.
6. The control device according to claim 1, wherein in the case that the determination has been made to permit the operation of the mist collector, the collector control unit stops driving the mist collector after a predetermined time period has elapsed from completion of the machining on the workpiece.
7. The control device according to claim 1, further comprising an alarm output unit configured to output an alarm in a case that an abnormality has occurred in the machine tool, wherein, in a case

that the alarm is output while the operation of the mist collector is ON, the collector control unit stops the operation of the mist collector.

8. A machine tool comprising the control device according to claim 1.

9. The machine tool according to claim 8, further comprising a sub-control device configured to control the mist collector instead of the collector control unit in a case that the control device is stopped.

10. A control method for controlling by a computer a machine tool comprising a mist collector configured to collect mist in a machining area, the machine tool being configured to machine a workpiece within the machining area, the control method comprising: an acquisition step of acquiring a type of a tool used for performing machining on the workpiece; a determination step of determining whether to permit or prohibit an operation of the mist collector based on the type of the tool; and a collector control step of turning ON the operation of the mist collector in a case that a determination has been made to permit the operation of the mist collector, and turning OFF the operation of the mist collector in a case that a determination has been made to prohibit the operation of the mist collector.